

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

**SP7 - General Insurance Reserving and Capital
Modelling**

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) ABC, an insurance company started its operations in 2020. It primarily writes Home Insurance policies and has been earning underwriting profits in all the years since inception. The company had entered into a quota share reinsurance arrangement for its entire portfolio when it began operations in 2020, which it has kept unchanged until 2024.

- i) State possible reasons why the company has kept its reinsurance arrangement unchanged from inception in 2020 to 2024? (4)

It is now 2025 and the company just concluded its annual business planning exercise. During this, the annual reinsurance program was evaluated and it was decided to revamp the structure for calendar year 2025.

- ii) What could be the reasons for the company to change its reinsurance structure for 2025? (5)

It was decided that the company will optimise its reinsurance program via XoL covers for its large losses. The company has received Excess of loss (Risk XL) reinsurance quotations from two reinsurers.

Reinsurer A:

XoL cover of ₹30 lakhs xs ₹10 lakhs with a 20% rate on line. It has one reinstatement at 250% cost, pro-rated to the amount and period reinstated. Additionally, it has a cover of ₹15 lakhs xs ₹40 lakhs at a 15% rate on line with no reinstatement.

Reinsurer B:

XoL cover of ₹35 lakhs xs ₹8 lakhs with a 40% rate on line, one reinstatement at 100% cost.

- iii) The company is trying to choose between the 2 options and decided to evaluate based on the losses it incurred during 2023 and 2024. Determine which reinsurance option the insurer should choose for each of the below scenarios:
- a) Ground up losses of ₹42 lakhs and ₹20 lakhs on 1st April, 2023 and 1st October, 2023 respectively.
- b) Ground up loss of ₹35 lakhs on 1st July 2024.

(8)

[17 Marks]

Q.2) Insurer ABC has been writing a variety of property and casualty business in Country Y for several years. Its multi-year motor insurance product is a key differentiator in the market has achieved recent success.

- i) Describe how underwriting a multi-year motor insurance differs from a single year motor policy. (3)

Country Y's insurance regulator is drafting regulations to make some major overhauls to the existing accounting approach. The proposed regulation is intended to provide a more comprehensive, transparent and consistent accounting approach.

- Standardised approach for all insurance contracts, expected to improve comparability and transparency across companies.
- Reflecting market value of assets and liabilities rather than the historical or book value.
- Recognition of profits as and when the services are delivered rather than immediately. Losses are however recognised immediately.
- More granular assumption setting and disclosure requirements.

- ii) What are the possible benefits of these regulatory changes? (6)

ABC insurance has employed you as a consulting actuary to assess the implications of the new accounting proposal.

- iii) Describe reasons why the company's financials may be worse off under the new regime. (7)
[16 Marks]

Q.3)

- i) Define Market Risk. (1)
- ii) List 4 sources of market risk. (2)
- iii) What are the factors to consider when modelling market risk. (4)

A General Insurance company writes a variety of short and long tailed lines of business. The company uses an internal model for SCR (Solvency Capital Requirement) calculation. It has been reinsuring a significant portion of its portfolios with a specific reinsurer for many years and has decided to switch to a different reinsurer, who is offering more favourable terms.

- iv) Suggest possible reasons why this might lead to increase in capital charge. (6)
- v) State considerations as Capital Modelling Actuary while modelling this change in the capital calculation. (6)

[19 Marks]

Q.4)

- i) State the differences between IBNR and IBNER? (1)

The actuarial team of Insurer QRS has been working on IBNR reserving for its home insurance business.

The team has obtained the below reported cumulative claims count triangle by accident year

Accident Year	1	2	3	4	5	6
2018	200	350	710	950	1000	1000
2019	250	550	850	1050	1100	
2020	150	400	800	950		
2021	350	900	1600			
2022	650	1000				
2023	250					

The team uses a Basic Chain ladder method to calculate ultimate claims count as at end of 2023.

- ii) Calculate the ultimate claim count for each accident year. (5)

It is expected that each claim would result in an average ultimate payout of ₹200,000 and the team assumes this is applicable for all accident years. The reported claim amount for each accident year as at end of 2023 has been obtained as below:

Accident Year	Reported Claim Amount (in 000s)
2018	1,60,000
2019	1,76,000
2020	1,52,000
2021	2,88,000
2022	1,90,000
2023	50,000

- iii) Using the above information, calculate the total IBNR reserve and split it into IBNER and pure IBNR components. (8)
- iv) Suggest ways to improve the above calculation. (2)

The underwriter expected a loss ratio of 85% when pricing the product and it remained the same for all years. Below are the Earned Premium by accident year.

Accident Year Earned Premium Amount (in 000s)

2020	2,50,000
2021	4,00,000
2022	5,00,000
2023	2,50,000

- v) Ignoring reinsurance, calculate the Ultimate loss ratio for each of the above accident years (2020-2023) using the Bornhuetter-Ferguson method. (6)
- vi) You have been tasked to communicate the results to the senior management of the insurer QRS.. Describe the considerations you should keep in mind while communicating the results, including but not limited to, the reasons for variability in results under different methods calculated above. (6)

[28 Marks]

Q.5) You are working in the actuarial team responsible for reserving, focusing primarily on retail lines of business. Your team currently uses average cost per claim method for IBNR estimation.

State the potential impact on IBNR estimate and adjustments you would make to the methodology in the below scenarios:

- i) Company decides to protect its home insurance portfolio with a stop loss reinsurance arrangement for the first time. (1.5)
- ii) Company had a waiting period clause in their Personal accident policy, that has now been removed. (1.5)
- iii) A new regulation is passed to remove discovery period for all products, including those already underwritten. (1.5)
- iv) Machine learning based claims investigation software that was recently adopted, that can potentially improve claims handling speed at a lower cost. (1.5)

[6 Marks]

Q.6)

- i) What are the types of claim inflation affecting motor insurance? (2)

Country X has a large motor insurance market and many insurance companies specialize in this line of business. Recently, the country's apex court has passed a landmark judgement for a motor third party accident claim. The key feature of this judgement is a revision in liability calculation, translating to increase in payout to the victim. Over and above this, an interest is awarded for delay in claim settlement since accident. This is now being considered as a landmark legal judgement by few other courts when settling similar cases.

You are an actuary at ABC insurance which writes sizable motor insurance business and has a dedicated team to handle third party claims since many years. You have been tasked by the management to ascertain the impact of these recent developments on the claim reserves and recommend a way forward.

- ii) Suggest analyses you would carry out to ascertain the financial impact on the company's claim reserves? (7)
- iii) Recommend different steps the company could take to reduce this impact. (5)

[14 Marks]
